

Chapter 4: Model Checking in the Cox Regression Model

Modelling Survival Data in Medical Research

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Part 2

1. Identifying Influential Observations

- ▷ Delta-betas
- ▷ Likelihood displacement
- ▷ $|\ell_{\max}|$ diagnostic

2. Testing Proportional Hazards

- ▷ Log-cumulative hazard plot
- ▷ Schoenfeld residual plot
- ▷ Grambsch-Therneau test
- ▷ Time-dependent variables

3. Q&A

Why Influence Matters

An **influential observation** is one whose removal substantially changes the model fit or conclusions.

Examples of influence in practice:

- ▶ One patient changes whether a treatment effect is significant or not
- ▶ One observation shifts the hazard ratio from < 1 to > 1
- ▶ One patient drives an unusual estimated coefficient

Challenge with the Cox model:

Unlike linear regression, we cannot simply express the log-likelihood as a sum of independent terms. Removing one observation changes the risk sets for all later death times — so exact calculations require re-fitting n separate models.

Solution: approximations based on score residuals.

Three diagnostics

- 1 **Delta-betas** $\Delta_i \hat{\beta}_j$: influence on individual parameter estimates
- 2 **Likelihood displacement** LD_i : influence on the overall log-likelihood
- 3 $|\ell_{\max}|$ **diagnostic**: influence on the entire set of parameter estimates

Delta-Betas: Influence on Individual Parameters

Definition

$\Delta_i \hat{\beta}_j \approx \hat{\beta}_j - \hat{\beta}_{j(i)}$ is the approximate change in $\hat{\beta}_j$ when observation i is omitted.

Vector of delta-betas for observation i : $\Delta_i \hat{\beta} = \mathbf{r}'_{Ui} \widehat{\text{var}}(\hat{\beta})$

How to use:

- ▶ Compute $\Delta_i \hat{\beta}_j$ for all i and each explanatory variable j
- ▶ Make an **index plot** of delta-betas: plot $\Delta_i \hat{\beta}_j$ vs. observation number
- ▶ Large $|\Delta_i \hat{\beta}_j| \Rightarrow$ observation i strongly influences $\hat{\beta}_j$
- ▶ Can also plot vs. rank order of survival times

Standardised delta-beta:

$$\frac{\Delta_i \hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)}$$

Interpreted as: the change in the Wald test statistic $\hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$ when observation i is removed.

This shows whether removing one observation would change whether $\hat{\beta}_j$ is significant.

Likelihood Displacement: Influence on Overall Fit

Approximate likelihood displacement (Pettitt & Bin Daud, 1989)

$$LD_i \approx 2 \left[\log L(\hat{\beta}) - \log L(\hat{\beta}_{(i)}) \right] \approx \mathbf{r}'_{Ui} \widehat{\text{var}}(\hat{\beta}) \mathbf{r}_{Ui}$$

The approximate change in $-2 \log \hat{L}$ when observation i is omitted from the fit.

Properties:

- ▶ A single number per observation (not per variable)
- ▶ Large $LD_i \Rightarrow$ observation i has a large effect on the entire model
- ▶ Index plot or plot vs. rank order of survival times
- ▶ Do **not** plot against explanatory variables (deterministic pattern even when model is correct)

Relation to score residuals:

$LD_i = \mathbf{r}'_{Ui} \widehat{\text{var}}(\hat{\beta}) \mathbf{r}_{Ui}$ is the diagonal element of the $n \times n$ matrix $\mathbf{B} = \mathbf{U} \widehat{\text{var}}(\hat{\beta}) \mathbf{U}'$, where $\mathbf{U}$ is the matrix of score residuals.

So all three diagnostics in this section come from the same score residual matrix.

The $|\ell_{\max}|$ Diagnostic

Definition

Let $\mathbf{B} = \mathbf{U}\widehat{\text{var}}(\hat{\boldsymbol{\beta}})\mathbf{U}'$ (the $n \times n$ influence matrix). The $|\ell_{\max}|$ **diagnostic** is formed from the **eigenvector corresponding to the largest eigenvalue of $\mathbf{B}$** , standardised to unit length.

Why the largest eigenvector?

The direction of greatest variation in the influence matrix points toward the observations that have the most effect on the set of parameter estimates jointly.

How to use:

- ▶ Plot $|\ell_{\max}|_i$ vs. index, rank order, or covariate values
- ▶ Cases where $|\ell_{\max}|_i^2$ accounts for a large share of 1.0 are most influential

Advantages over delta-betas:

- ▶ Single value per observation (not per variable)
- ▶ Detects observations that jointly influence the parameter vector even when no single $\Delta_i \hat{\beta}_j$ is extreme
- ▶ Plotting against covariates is valid (no deterministic pattern under a correct model)

Example 4.7 — Kidney Data: Influence Results

Patient	LD_i	$ \ell_{\max} _i$
1	0.033	0.161
2	0.339	0.309
3	0.005	0.068
4	0.338	0.621
5	0.050	0.104
6	0.019	0.058
7	0.136	0.291
8	0.027	0.054
9	0.133	0.124
10	0.035	0.193
11	0.061	0.264
12	0.043	0.224
13	0.219	0.464

Key findings:

- ▶ Patients **2, 4, and 13** are most influential
- ▶ $|\ell_{\max}|_4 = 0.621$ is largest: patient 4 most affects the parameter vector overall
- ▶ Patients 2, 4, 13 together account for $(0.309^2 + 0.621^2 + 0.464^2)/(1.0) \approx 70\%$ of the total
- ▶ Delta-beta analysis (Ex. 4.6) confirmed patients 2 and 4 most affect the Sex coefficient

Action taken: omitting each of these three patients changes the linear component, but the changes are within one standard error for each coefficient; there is no reason to exclude them.

What to Do with Influential Observations

Once identified, influential observations should be investigated, but not automatically removed.

Step 1: Check for data errors

- ▶ Transcription errors in survival time or covariate values are common
- ▶ If an error is found, correct it and refit the model
- ▶ If correction is impossible and the value is unrealistic, removal may be justified

Step 2: Assess practical impact

- ▶ Compare key estimates (HR, median survival) with and without the influential observation
- ▶ If the difference is negligible, retain the observation

Step 3: Report transparently

- ▶ If the impact is non-trivial, report analyses both with and without the observation
- ▶ Consult subject matter experts
- ▶ Do **not** silently drop influential observations without justification

Golden rule

Influential observations should **never** be deleted simply because they are influential. The question is always: *are they valid data?*

The Proportional Hazards Assumption

The Cox model assumes that the hazard ratio between any two individuals is **constant over time**:

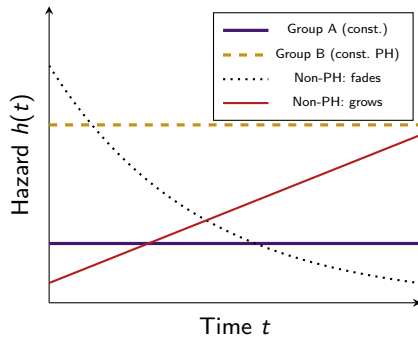
$$\frac{h_i(t)}{h_k(t)} = \frac{\exp(\hat{\beta}' \mathbf{x}_i)}{\exp(\hat{\beta}' \mathbf{x}_k)} = e^{\hat{\beta}'(\mathbf{x}_i - \mathbf{x}_k)} \quad (\text{does not depend on } t)$$

When does PH fail?

- ▶ Treatment benefit that fades or strengthens over time
- ▶ An early "crossing" of survival curves
- ▶ A covariate whose effect changes after a threshold time

Three methods to check PH:

- 1 Log-cumulative hazard plot (graphical, before fitting)
- 2 Scaled Schoenfeld residual plot (graphical, after fitting)



The Log-Cumulative Hazard Plot

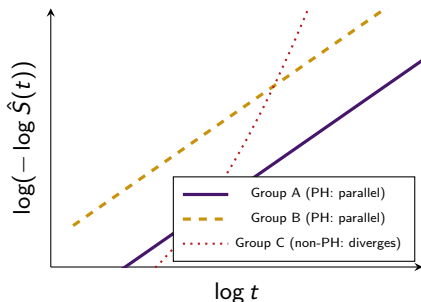
From the Cox model: $H_i(t) = e^{\beta' \mathbf{x}_i} H_0(t)$

Taking log: $\log H_i(t) = \beta' \mathbf{x}_i + \log H_0(t)$

Key result: for any two individuals i and k , the difference $\log H_i(t) - \log H_k(t) = \beta'(\mathbf{x}_i - \mathbf{x}_k)$ is *constant in time*. So log-cumulative hazard curves must be **parallel** under PH.

How to construct:

- 1 Group patients by levels of a factor (or categorical version of a continuous variable)
- 2 Compute KM estimate $\hat{S}(t)$ for each group
- 3 Plot $\log \hat{H}(t) = \log(-\log \hat{S}(t))$ vs. $\log t$
- 4 Parallel curves $\Rightarrow$ PH holds
- 5 Crossing or diverging curves $\Rightarrow$ PH violated



Scaled Schoenfeld Residuals: Testing PH After Fitting

Grambsch-Therneau (1994) result

The expected value of the i th scaled Schoenfeld residual for variable X_j is approximately:

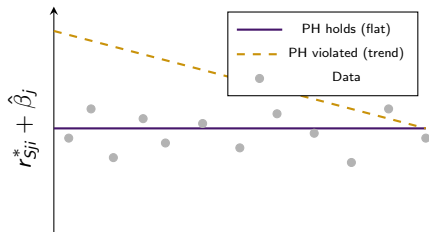
$$E(r_{Sji}^*) \approx \beta_j(t_i) - \hat{\beta}_j$$

where $\beta_j(t)$ is a *time-varying* version of the coefficient. If PH holds, $\beta_j(t) = \hat{\beta}_j$ (constant), so $E(r_{Sji}^*) \approx 0$ for all t .

Diagnostic plot:

Plot $r_{Sji}^* + \hat{\beta}_j$ (or just r_{Sji}^*) vs. survival time t_i .

- ▶ A *horizontal* scatter $\Rightarrow$ PH holds for X_j
- ▶ A trend (increasing or decreasing) $\Rightarrow$ PH may be violated
- ▶ Superimpose a smoother to aid interpretation



The Grambsch-Therneau Test (zph Test)

Formal test of H_0 : $\beta_j(t) = \text{constant}$ (PH holds for X_j).

The test fits a linear regression of r_{Sji}^* on time and tests whether the slope ν_j is zero.

Test statistic for variable X_j (1 d.f.)

$$\text{Test}_j = \frac{\left[\sum_{i=1}^d (\tau_i - \bar{\tau}) r_{Sji}^* \right]^2}{\widehat{\text{var}}(\hat{\beta}_j) \cdot \sum_{i=1}^d (\tau_i - \bar{\tau})^2} \sim \chi_1^2 \text{ under } H_0$$

Global test across all p variables (p d.f.)

$$\text{Global test} = \frac{(\boldsymbol{\tau} - \bar{\tau}\mathbf{1})' \mathbf{S} \widehat{\text{var}}(\hat{\boldsymbol{\beta}}) \mathbf{S}' (\boldsymbol{\tau} - \bar{\tau}\mathbf{1})}{d^{-1} \sum_i (\tau_i - \bar{\tau})^2} \sim \chi_p^2$$

τ_i = i th event time, $\bar{\tau}$ = mean event time, $\mathbf{S}$ = matrix of Schoenfeld residuals.

Can use $\log \tau_i$, rank order, or $\hat{S}(\tau_i)$ instead of τ_i as the time scale.

Adding a Time-Dependent Variable to Test PH

A direct formal test: add $X_j \times t$ to the model and test whether its coefficient is zero.

For a binary treatment variable X_1 :

Model without PH assumption:

$$h_i(t) = \exp(\beta_1 x_{1i} + \beta_2 x_{1i} t) h_0(t)$$

Relative hazard of new vs. standard treatment:

$$e^{\beta_1 + \beta_2 t}$$

- ▶ $\beta_2 = 0$: PH holds
- ▶ $\beta_2 < 0$: treatment benefit increases over time
- ▶ $\beta_2 > 0$: treatment benefit decreases over time

Testing:

Test $H_0 : \beta_2 = 0$ using the LRT (compare $-2 \log \hat{L}$ with and without $X_1 t$).

Practical notes:

- ▶ Use $X_1(t - t_0)$ to reduce correlation between $\hat{\beta}_1$ and $\hat{\beta}_2$
- ▶ t_0 = mean or median survival time is a natural choice
- ▶ Use $\log t$ instead of t when survival times are large (avoids numerical issues)
- ▶ These models are fitted in R using `tt()` or counting process format (Chapter 8)

Example 4.11/4.12 — Kidney Data: PH Tests

Fitted model: $\hat{h}_i(t) = e^{0.030 \text{ Age}_i - 2.711 \text{ Sex}_i} \hat{h}_0(t)$

Grambsch-Therneau test (Section 4.4.3):

Variable	χ^2 (1 d.f.)	P
Age	0.811	0.368
Sex	0.224	0.636
Global (2 d.f.)	0.811	0.667

No evidence against PH for either variable.

Time-dependent variable test (Section 4.4.4):

Added term	$\Delta(-2 \log \hat{L})$	P
$T_{\text{age}} = \text{Age} \times t$	2.462	0.117
$T_{\text{sex}} = \text{Sex} \times t$	0.364	0.546

Confirms no time-dependence in either coefficient.

Three complementary methods (graphical, formal test, time-dependent variable) all reach the same conclusion: the PH assumption is valid for these data.

Recommendations: Routine Diagnostics

For every fitted Cox model:

1. Overall fit

- ▶ Plot **deviance residuals** vs. risk score $\hat{\eta}_i$
- ▶ Identify any obvious outliers ($|r_{Di}| > 2$)
- ▶ Can supplement with Cox-Snell cumulative hazard plot

2. Functional form

- ▶ Plot martingale residuals (null model) vs. each continuous covariate
- ▶ Superimpose LOWESS smoother
- ▶ A straight line $\Rightarrow$ linear term is appropriate

3. Influence

- ▶ If one or two parameters are of primary interest: use **delta-betas** and plot vs. rank order of survival times
- ▶ For overall influence: use $|\ell_{\max}|$ diagnostic and index plot

4. Proportional hazards

- ▶ Preliminary check: **log-cumulative hazard plot**
- ▶ After fitting: **scaled Schoenfeld residual plots**
- ▶ Formal test: **Grambsch-Therneau (zph)** test in R
- ▶ If suspected: add $X_j \times t$ term and test with LRT

No single diagnostic is definitive. Use multiple complementary tools.
A satisfactory model passes *all* checks, not just one.

Six types of residuals:

- ▶ Cox-Snell r_{Ci} : unit exponential under correct model
- ▶ Modified CS r'_{Ci} : corrects censored residuals
- ▶ Martingale $r_{Mi} = \delta_i - r_{Ci}$: observed minus expected deaths
- ▶ Deviance r_{Di} : symmetric transformation of r_{Mi}
- ▶ Schoenfeld r_{Sji} : weighted residuals at each death time
- ▶ Score r_{Uji} : influence building blocks

Model fit assessment:

- ▶ Cox-Snell plot: quick check, not reliable
- ▶ Deviance residual plots: outlier detection
- ▶ Null-model martingale: functional form

Influence:

- ▶ Delta-betas: influence on $\hat{\beta}_j$ individually
- ▶ Likelihood displacement LD_i : overall influence on log-likelihood
- ▶ $|\ell_{\max}|$: influence on entire parameter vector

Proportional hazards:

- ▶ Log-cumulative hazard plot: preliminary graphical check
- ▶ Scaled Schoenfeld plots: detect time-varying coefficients
- ▶ Grambsch-Therneau test: formal χ^2 test per variable + global
- ▶ Time-dependent $X_j \times t$: direct LRT test

Questions?

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